

Why it matters

Hospital discharge is a fragile medication transition. Changes can be omitted, duplicated, or altered across admission and discharge records.

The design principle

The AI proposes. Deterministic code verifies. The clinician decides.

System architecture



Evidence provenance is attached to every surfaced finding. When identity or transition context is insufficient, SentinelRx abstains instead of forcing a conclusion.

Reproducible evaluation

1.000	0.600	1.000	0.000
F1	Baseline F1	Abstention	Crash rate
120 cases	120 cases	90 cases	90 cases

Generated synthetic test conditions only - not clinical validation.

What judges can inspect

1 Safety Review

Potential omissions, additions, duplicates and dose changes.

2 Medication Timeline

Admission vs discharge medication view.

3 Evidence Provenance

Finding-to-FHIR resource traceability.

4 Counterfactual Lab

Change synthetic evidence and rerun the same engine.

5 Evaluation

Standard + failure-handling benchmarks.

Competition contribution

SentinelRx is a separate competition codebase built from lessons learned in the earlier FHIRGuard prototype, with new clinician workflow, provenance, counterfactual analysis and expanded evaluation.

Health & rigor

Synthetic FHIR cases have predefined ground truth. A transparent medication-set baseline provides a sanity comparator. Robustness cases test missing context, missing medication identity, missing dosage and malformed entries.

Safety boundary

Synthetic data only. Clinical decision-support prototype. No diagnosis, prescribing or autonomous treatment recommendation. A finding must show evidence provenance or the system should abstain.